SD5531 / TEAM 2
Yichen Zhang | Kecheng Zhu | Haolin Yang | James Eric Liu

Leopard Gecko Behavior Monitoring



Project 2 • Video Recording Module

Assignment 2
Project scope and technical exploration

A



PROJECT SCOPE

Purpose and wider project goal

Sustained evidence for student observation

1 Observation and research

Find target events, compare dates and develop questions using live views and recorded evidence.

2 Behavior and environment

Compare how geckos use the enclosure alongside layout, environmental conditions and operation records.

3 Care communication

As an extension, students select a clip, add an observation and ask a teacher to review it.

System scope

One observation record for each identifiable gecko

01	**Capture and retain**	Video and thermal-array acquisition, server storage and remote access.
02	**Identify and locate**	Explore pattern-based identity and locate the gecko in the image.
03	**Organize and share**	Associate records and analysis with time, region and the shared interface.

Individual identification remains subject to experimental validation.

Behavior observation

Activity and stillness

Location, movement and changes in activity.

Hide and region use

Entry and exit, hiding, and time spent in heating and humid areas.

Feeding, drinking and defecation

Timestamp candidate clips for students to replay and confirm.

The timeline connects every event to evidence that users can review.

Thermal observation

Surface temperature during valid visible periods

Associate thermal records with the gecko’s location, behavior and environmental temperature.

Validate measurement performance experimentally.
Retain gaps when measurement is not possible.



Information students can use

Spatial use

Dwell-time heatmaps and movement paths.

Behavior timeline

Time-stamped events linked to replayable clips.

Observation timing

Suggestions based on individual history, with past clips as references.

Teacher review / extension

Student-selected clips with notes for a teacher to review.

Delivery scope and boundaries

Within the proposed scope

- Capture and storage
- Identification and location
- Behavior information organization
- Data sharing and record alignment
- Camera count and placement
- Based on enclosure and team designs

Shared interface integration is planned.
A standalone demo depends on time.

Validation required

Pattern-based identification
Behavior recognition
Surface temperature measurement

Automatic anomaly detection and health diagnosis are outside the current deliverables.

Project validation

Can sustained records help students understand individual behavior?

Find opportunities to observe

Compare records and develop research questions

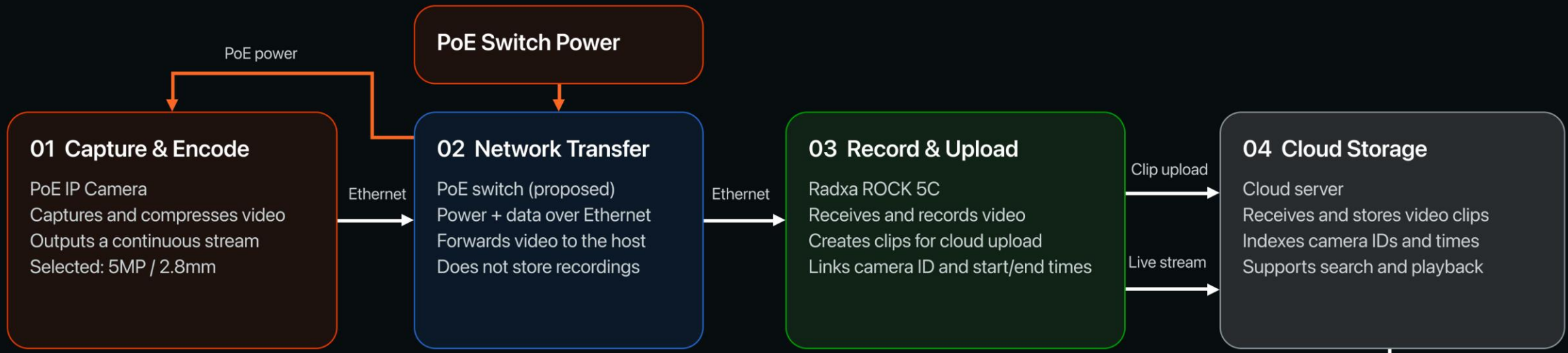
Use evidence in learning and care communication

B



TECHNICAL EXPLORATION

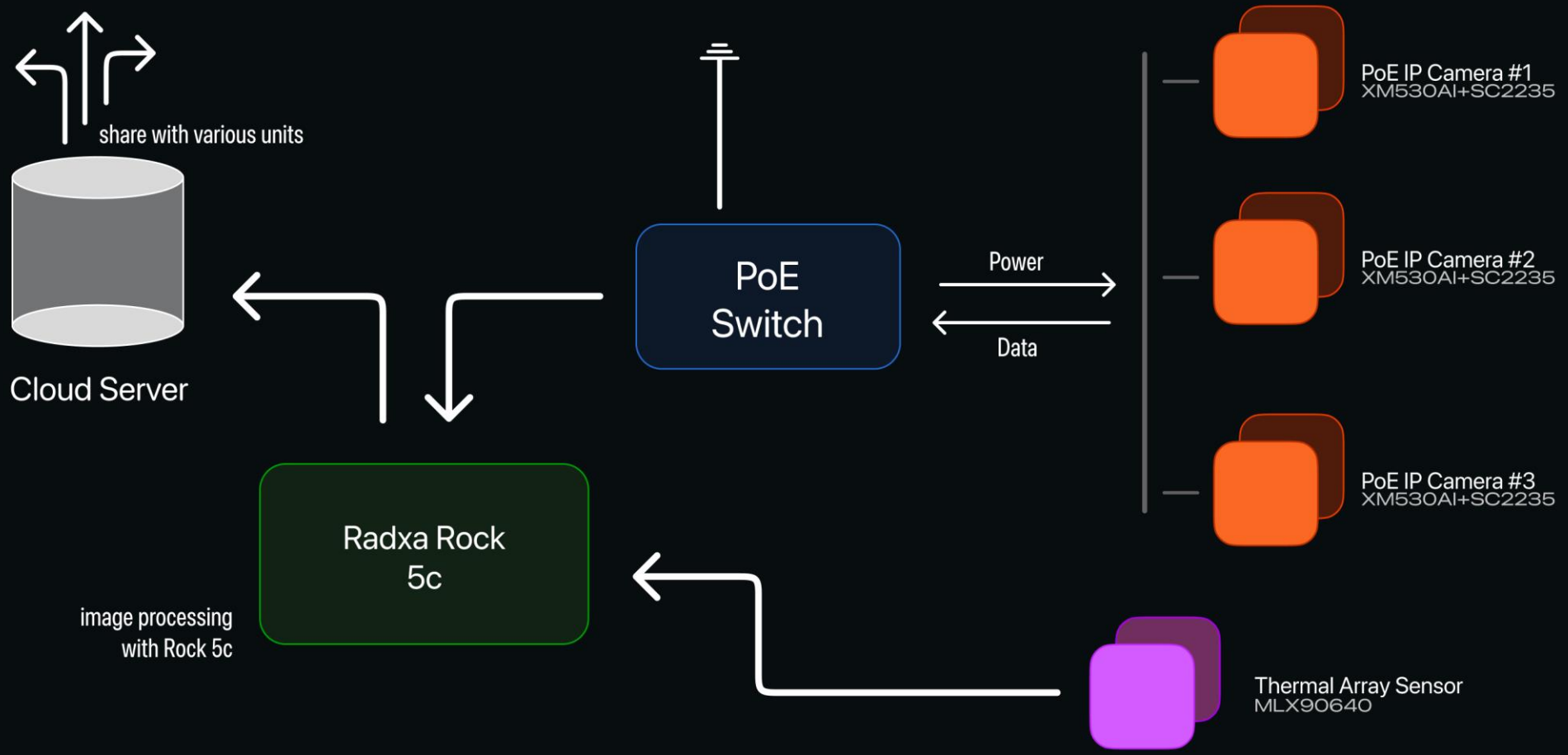
Behavior observation



How Video Clips Are Created and Stored

Camera streams video → ROCK 5C creates and uploads clips → Cloud stores and indexes clips → Students search and replay via the unified UI.

Data Transmission



Overall Hardware Selection



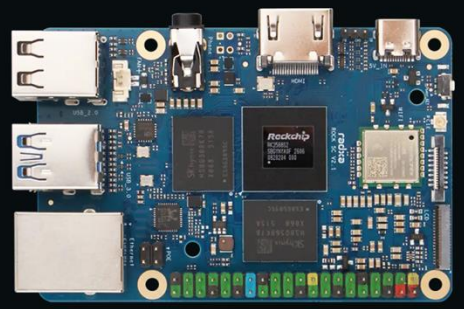
PoE IP Camera x N

XM530AI+SC2235



Thermal Array Sensor x N

MLX90640



Radxa Rock 5c x 1

Processing Board



5-Port 100 Mbps PoE x 1

Industrial version

Systematic Functions

1. Real-time Monitoring and Online Observing

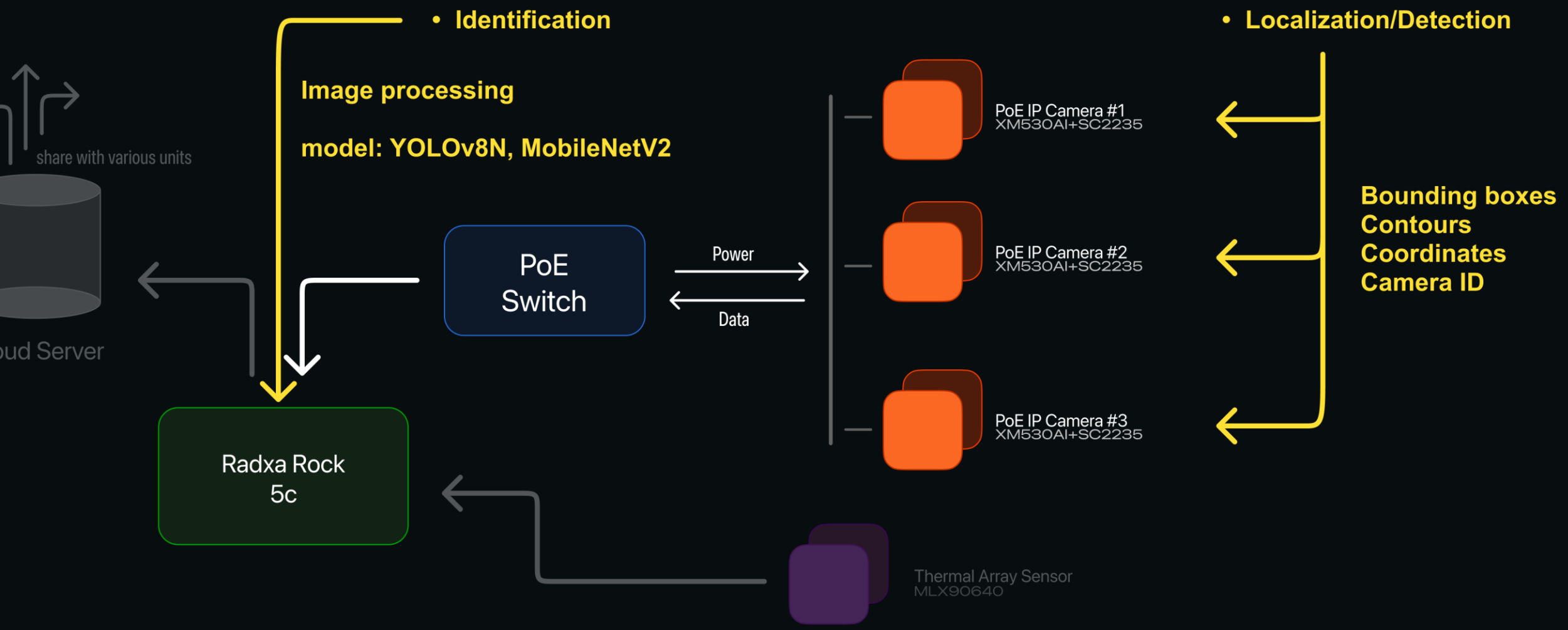


- general activity capture
- basking area

- water pond
- poop detection

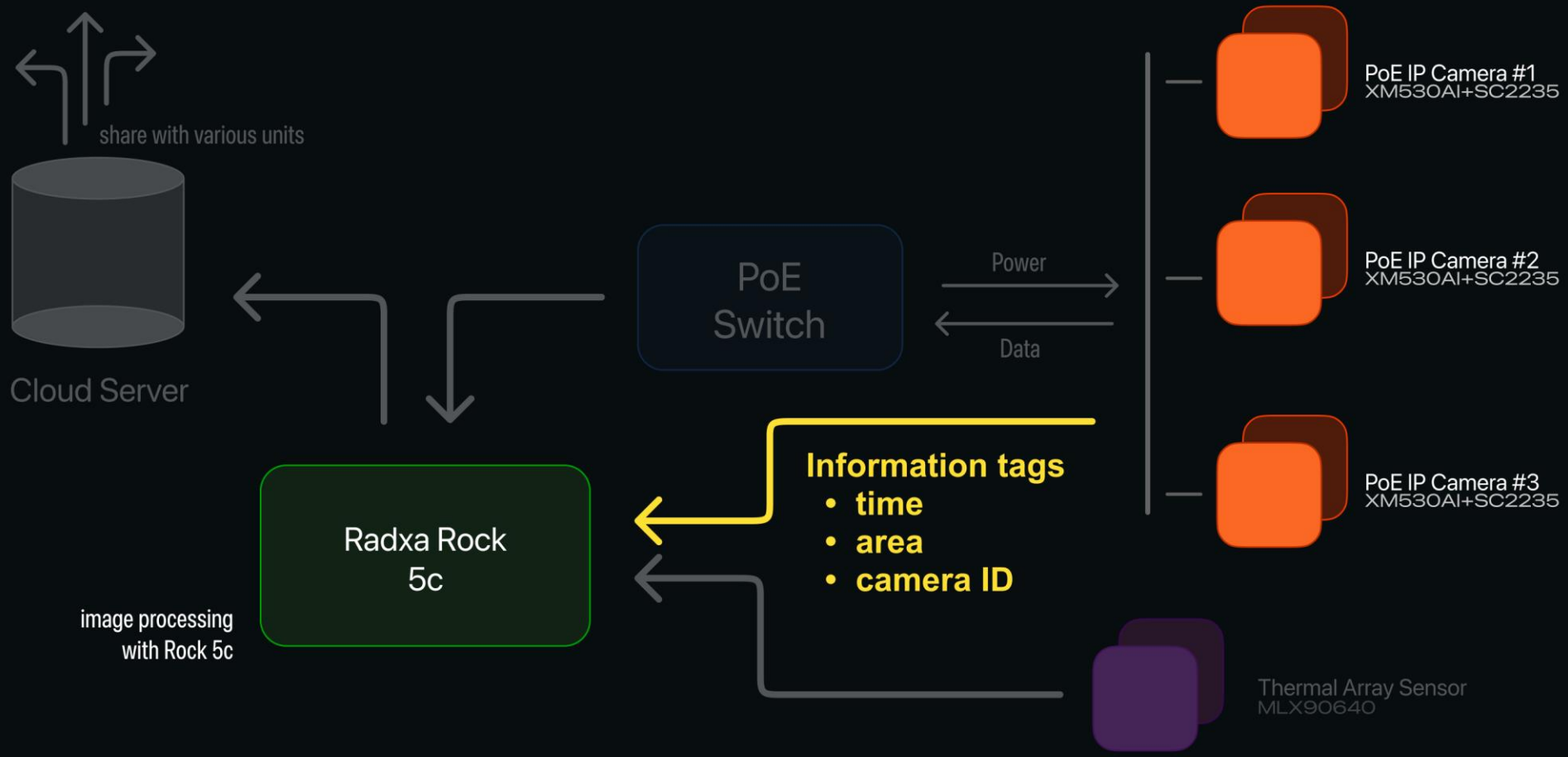
- hide

2. Individual Identification and Location Tracking

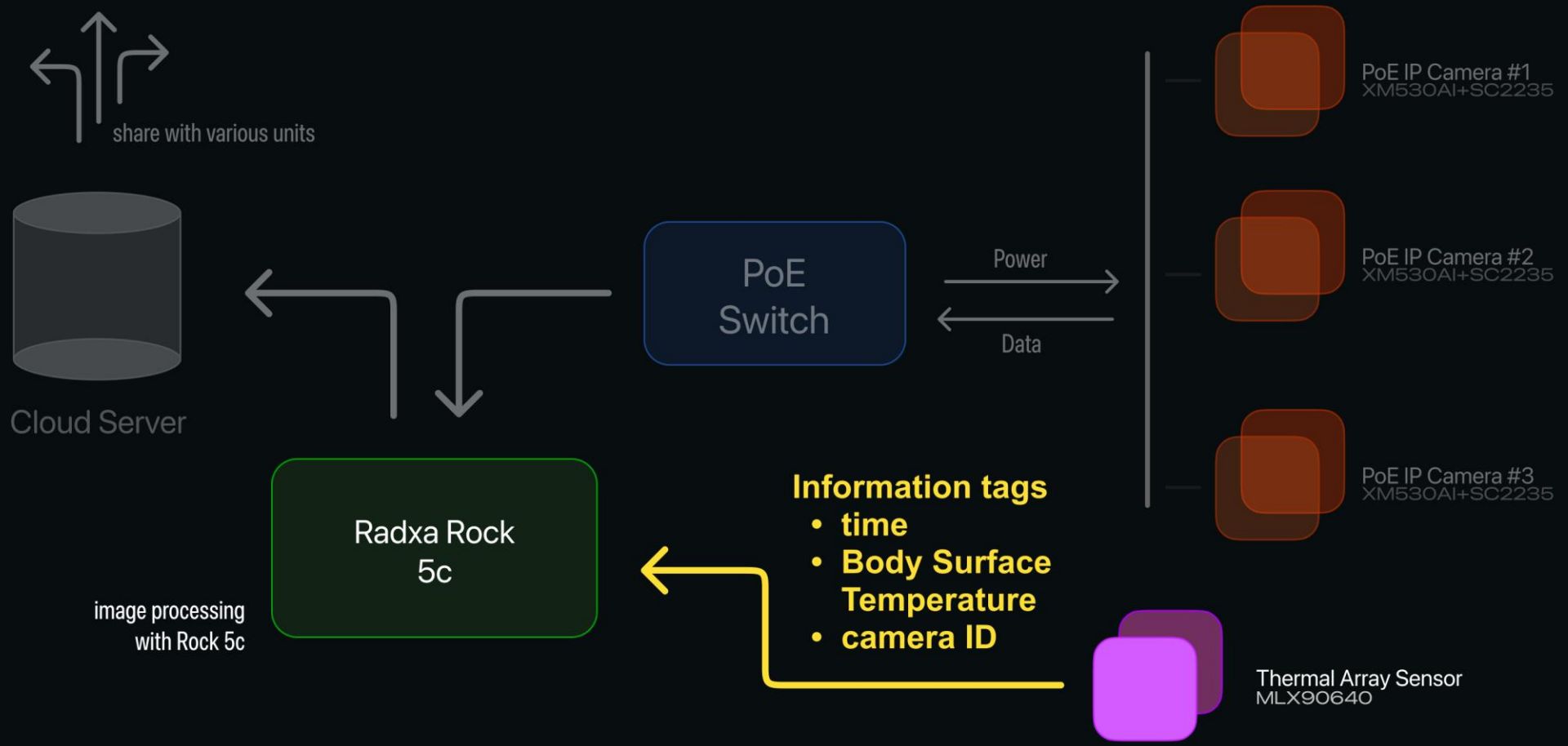


Behavior Monitoring

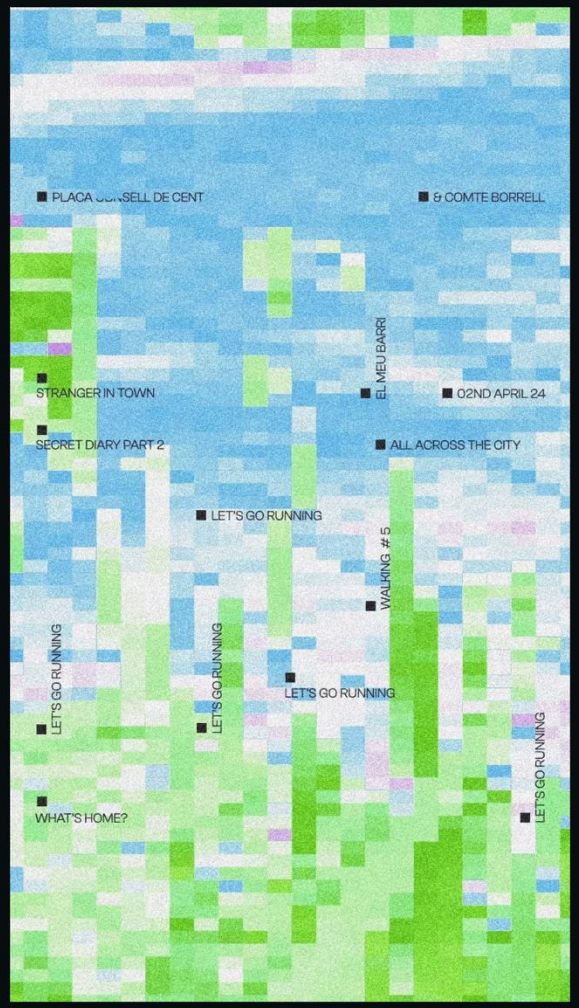
1. Organization of Daily Behavioral Information

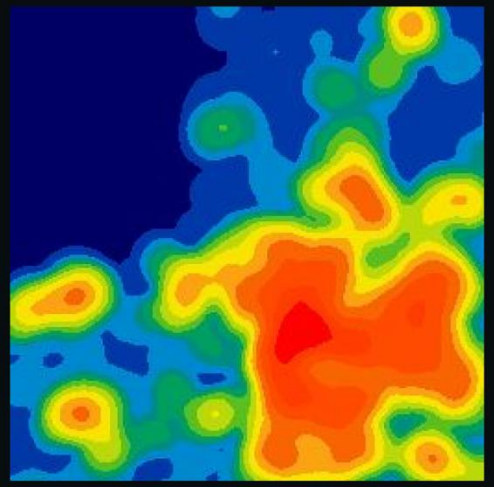


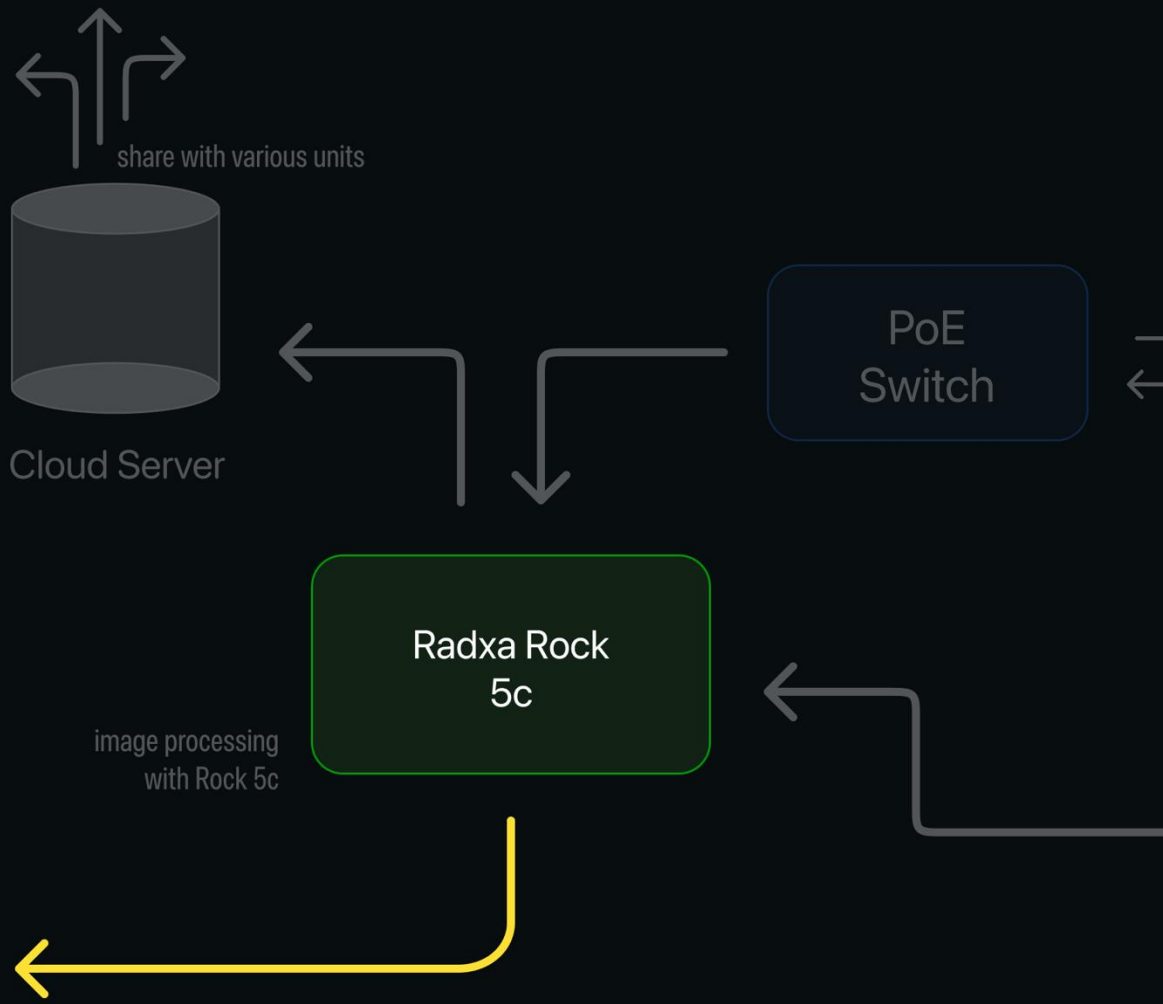
2. Body Surface Temperature Change



3. Visualization of activity patterns





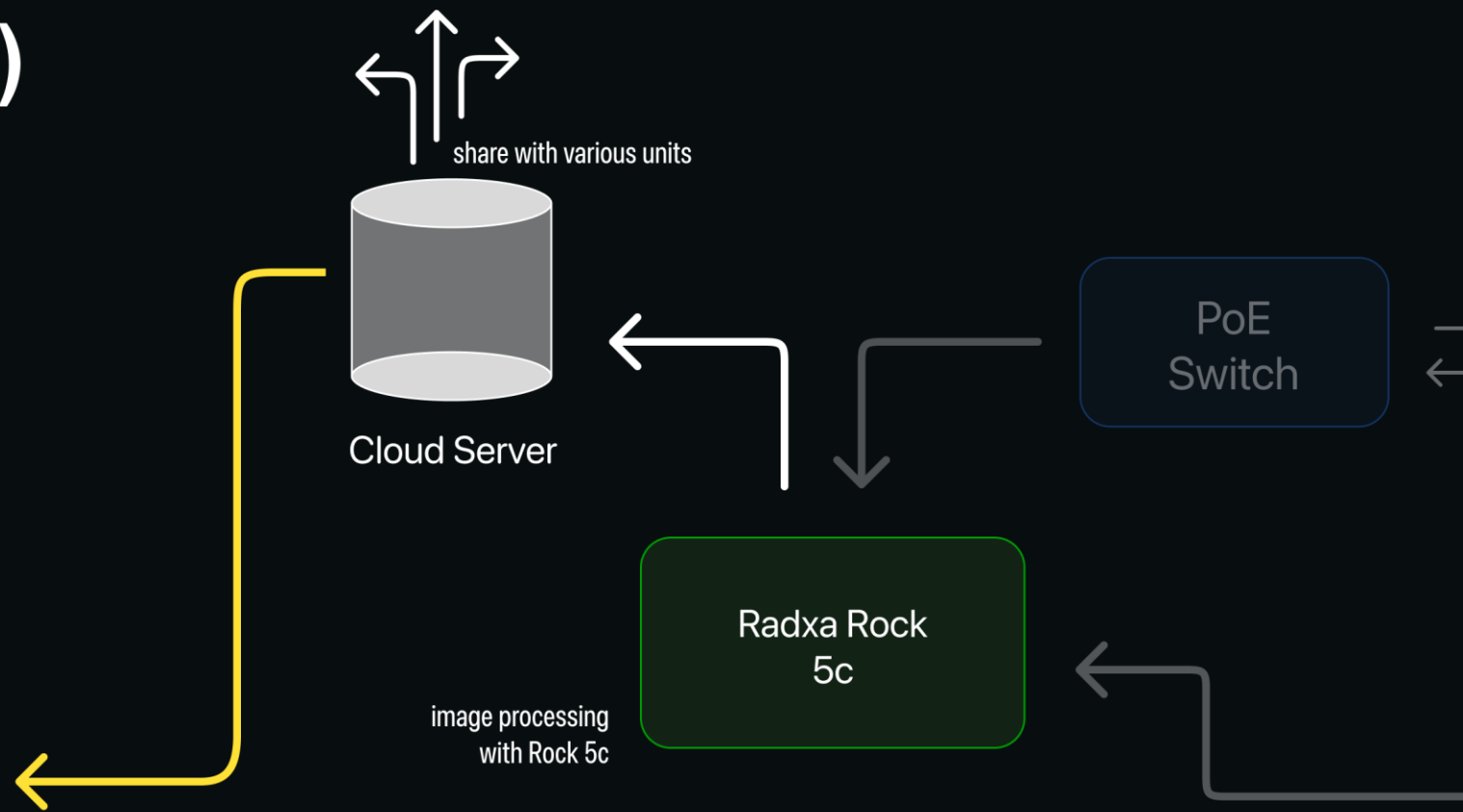


Heatmaps

- **Dwell time**
- **Activity duration**
- **Display movement and timelines**

Learning Support and Feedback (Extended)

1. Observation period recommendation
2. Anomalous Clues and Reporting



Basic ML models are trained for

- **providing time suggestions in observing a certain behavior**
- **identifying abnormal behaviors**

Team Work Distribution

1. Mechanical/Structural design, communicating with different groups for implementation —— **ZHANG Yichen, LIU Eric**
2. Hardwares: coding, functionality test on Rock 5c and soldering —— **ALL GROUP MEMBERS**
3. Softwares: Video clip labelling, Port sommunication, Final layout —— **YANG Haolin, ZHU Kecheng**
4. ML model training and deployment for image processing —— **YANG Haolin, ZHU Kecheng**

C



CROSS-TEAM COLLABORATION

Team inputs and outputs | 1 / 3

Partner team	Partner team provides	Our team provides	Proposed camera view
P1 Enclosure	Confirm dimensions, layout, access points, mounts and cable routes. Map blind spots together.	Movement paths and spatial heatmaps.	Top view of the main activity area. Add an angled view where obstructed.
P3 Hide	All entrances, internal structure, temperature, humidity and existing occupancy signals.	Entry and exit counts and dwell times.	Angled view of entrances, covering thresholds and entry / exit paths.

P = Project number. Camera views are proposals for testing and may be shared.

Team inputs and outputs | 2 / 3

Partner team	Partner team provides	Our team provides	Proposed camera view
P4 Basking / heating	Upper and lower structures, heat-source positions, on / off logs and measured temperatures.	Dwell records for each level.	Top view of the slate. Check whether the lower entrance is visible.
P5 Water / humidity	Bowl position, temperature and humidity sampling points, refill and humidification logs.	Visits and candidate drinking clips.	View from above and to the side, keeping the mouth visible above the bowl rim.

P = Project number. Camera views are proposals for testing and may be shared.

Team inputs and outputs | 3 / 3

Partner team	Partner team provides	Our team provides	Proposed camera view
P6 & P7 Environment / defecation	Confirm responsibilities first. Provide defecation areas, substrate, cleaning and lighting logs.	Before / after comparisons and event replay.	Fixed top view of the target substrate area.
P9 Feeding	Confirm responsibility for actual feeding. Provide feeding-area structure, feeding times and quantities.	Prey-capture clips and feeding clips confirmed by review.	Close angled top view covering food, mouth and approach paths.

P = Project number. Camera views are proposals for testing and may be shared.

Collaboration process

Stage	Joint action
01 Requirements	Each team names one contact and selects the 1–2 most important observation events.
02 Layout	Combine camera views and cable routes with P1. Identify blind spots and removable parts for cleaning.
03 Joint trial	Each team selects one event. Manually compare video with sensor records and correct occlusion or misclassification.
04 Integration	Standardize timestamps and region / individual IDs. Deliver heatmaps, activity distributions and event replay.

Shared agreements

Responsibilities	Functional teams own module data. Our team integrates video and activity timing. Confirm event labels together.
Record quality	Log invisible periods and stream interruptions. Confirm drinking and feeding through review.
Layout and thermal views	Update the layout after moving modules or cameras. If thermal arrays are used, give them a direct view without ordinary glass in between.
Coordination to confirm	Assign responsibility for app and algorithm integration.

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THANK YOU

